

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment-9

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Subject Name: Project Based Learning in Java

Subject Code: 22CSH-359

9.1.1 Aim: To demonstrate dependency injection using Spring Framework with Java-based configuration.

9.1.2 Objective:

Define Course and Student classes.

Use Configuration and Bean annotations to inject dependencies.

Load Spring context and print student details.

9.1.3 Code:

// Course.java

```
public class Course {
    private String courseName; private
    String duration;

    public Course(String courseName, String duration) { this.courseName =
        courseName;
        this.duration = duration;
    }

    public String getCourseName() { return courseName; } public
    String getDuration() { return duration; }

    @Override
    public String toString() {
        return "Course: " + courseName + ", Duration: " + duration;
    }
}
```

// Student.java

```
public class Student { private
    String name; private Course
    course;
```



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```
public Student(String name, Course course) {
    this.name = name;
    this.course = course;
}

public void showDetails() { System.out.println("Student: " +
    name); System.out.println(course);
}
} // AppConfig.java
import org.springframework.context.annotation.*;

@Configuration
public class AppConfig { @Bean
    public Course course() {
        return new Course("Java", "3 months");
    }

    @Bean
    public Student student() {
        return new Student("Aman", course());
    }
} // MainApp.java
import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.AnnotationConfigApplicationContext;

public class MainApp {
    public static void main(String[] args) { ApplicationContext
        context = new
        AnnotationConfigApplicationContext(AppConfig.class); Student
        student = context.getBean(Student.class);
        student.showDetails();
    }
}
```

Output:

```
Student: Aman
Course: Java, Duration: 3 months
```

9.2.1 **Aim:** To perform CRUD operations on a Student entity using Hibernate ORM with MySQL.

Objective: Define Course and Student classes.

Use Configuration and Bean annotations to inject dependencies. Load Spring context and print student details.

9.2.2 **Code:**

```
<hibernate-configuration>
  <session-factory>
    <property name="hibernate.connection.driver_class">com.mysql.cj.jdbc.Driver</property>
    <property name="hibernate.connection.url">jdbc:mysql://localhost:3306/testdb</property>
    <property name="hibernate.connection.username">root</property>
    <property name="hibernate.connection.password">password</property>
    <property name="hibernate.dialect">org.hibernate.dialect.MySQL8Dialect</property>
    <property name="hibernate.hbm2ddl.auto">update</property>
    <mapping class="Student"/>
  </session-factory>
</hibernate-configuration> import
```

```
javax.persistence.*;
```

Entity

```
public class Student { Id
    GeneratedValue(strategy = GenerationType.IDENTITY) private int id;
    private String name; private
    int age;
```

```
    public Student() {}
    public Student(String name, int age) { this.name =
        name;
        this.age = age;
    }
    // Getters, setters, toString
}
import org.hibernate.SessionFactory; import
org.hibernate.cfg.Configuration;
```

```
public class HibernateUtil {
    private static final SessionFactory sessionFactory;

    static {
```



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```
        sessionFactory = new Configuration().configure().buildSessionFactory();
    }

    public static SessionFactory getSessionFactory() { return
        sessionFactory;
    }
}

import org.hibernate.*; public

class MainCRUD {
    public static void main(String[] args) {
        Session session = HibernateUtil.getSessionFactory().openSession();

        // Create
        Transaction tx = session.beginTransaction();
        Student s1 = new Student("Aman", 22);
        session.save(s1);
        tx.commit();

        // Read
        Student student = session.get(Student.class, 1);
        System.out.println(student);

        // Update
        tx = session.beginTransaction();
        student.setAge(23);
        session.update(student); tx.commit();

        // Delete
        tx = session.beginTransaction(); session.delete(student);
```



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```
tx.commit();  
  
session.close();  
}  
}
```

9.2.3 Output:

```
Student{id=1, name='    ', age=22}  
Updated age to 23  
Deleted student with id 1
```



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9.3.1 Aim: To implement a banking system using Spring and Hibernate that ensures transaction consistency during fund transfers.

Objective:

Integrate Spring + Hibernate.

Handle transactions atomically (rollback on failure). Demonstrate success and failure cases.

Code:

```
import javax.persistence.*;
```

Entity

```
public class Account { @Id
    private int accountId; private
    String holderName; private double
    balance;
```

```
    // Constructors, getters, setters
}
```

```
import javax.persistence.*; import
java.util.Date;
```

@Entity

```
public class BankTransaction { @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY) private int
    txnId;
    private int fromAcc; private
    int toAcc; private double
    amount;
    private Date txnDate = new Date();
```

```
    // Constructors, getters, setters
}
```

```
import org.hibernate.*;
import org.springframework.transaction.annotation.Transactional;
```

```
public class BankService {
    private SessionFactory sessionFactory;
```



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```
public BankService(SessionFactory sessionFactory) { this.sessionFactory
    = sessionFactory;
}

@Transactional
public void transferMoney(int fromId, int toId, double amount) { Session session =
    sessionFactory.getCurrentSession();

    Account from = session.get(Account.class, fromId); Account to =
    session.get(Account.class, toId);

    if (from.getBalance() < amount) {
        throw new RuntimeException("Insufficient Balance");
    }

    from.setBalance(from.getBalance() - amount);
    to.setBalance(to.getBalance() + amount);

    session.update(from);
    session.update(to);

    BankTransaction txn = new BankTransaction(fromId, toId, amount); session.save(txn);
}
}

@Configuration
@EnableTransactionManagement public
class AppConfig {

    @Bean
    public DataSource dataSource() {
        DriverManagerDataSource ds = new DriverManagerDataSource();
        ds.setDriverClassName("com.mysql.cj.jdbc.Driver");
        ds.setUrl("jdbc:mysql://localhost:3306/testdb"); ds.setUsername("root");
        ds.setPassword("password");
        return ds;
    }

    @Bean
    public LocalSessionFactoryBean sessionFactory() { LocalSessionFactoryBean lsf = new
        LocalSessionFactoryBean(); lsf.setDataSource(dataSource());
        lsf.setPackagesToScan("your.package");
```



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```
Properties props = new Properties();
props.put("hibernate.dialect", "org.hibernate.dialect.MySQL8Dialect");
props.put("hibernate.hbm2ddl.auto", "update"); lsf.setHibernateProperties(props);
return lsf;
}

@Bean
public HibernateTransactionManager transactionManager(SessionFactory sf) { return new
    HibernateTransactionManager(sf);
}

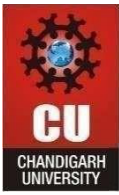
@Bean
public BankService bankService(SessionFactory sf) { return new
    BankService(sf);
}
}

public class MainApp {
    public static void main(String[] args) {
        AnnotationConfigApplicationContext ctx = new
        AnnotationConfigApplicationContext(AppConfig.class);
        BankService service = ctx.getBean(BankService.class);

        try {
            service.transferMoney(101, 102, 500);
            System.out.println("Transaction Successful!");
        } catch (Exception e) {
            System.out.println("Transaction Failed: " + e.getMessage());
        }
        ctx.close();
    }
}
```

OUTPUT

```
Transaction Successful!
OR
Transaction Failed: Insufficient Balance
```

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Learning Outcomes

- Demonstrated Dependency Injection using Spring with Java-based configuration via `@Bean` and `@Configuration`.
- Performed CRUD operations on Student entity using Hibernate ORM with MySQL database.
- Integrated Spring + Hibernate for seamless object-relational mapping and dependency management.
- Implemented transaction management using `@Transactional` to ensure atomicity in fund transfers.
- Handled transaction failures and rollbacks (e.g., insufficient balance) to maintain data consistency.